Lectures on Monte Carlo Theory

Chapter 1: Introduction to Monte Carlo

Based on Leskälä & Vihola

Chapter Outline

1.1 What Are Monte Carlo Methods? I

Definition — Monte Carlo (MC) Methods

Monte Carlo methods are a broad family of computational algorithms that rely on *repeated random (or pseudo-random) sampling* to obtain numerical results. They are used to solve problems in numerical integration, optimisation, simulation of stochastic systems, Bayesian statistics, and many other fields where exact analytical solutions are unavailable or too costly to compute.

The name "Monte Carlo" was coined by Metropolis and Ulam in 1949, after the famous casino in the Principality of Monaco, because the method resembles repeated gambling experiments: you play the same random game many times and record the outcomes. The approach was developed at Los Alamos National Laboratory in the 1940s by Stanislaw Ulam, John von Neumann, and Nicholas Metropolis, originally to simulate how neutrons diffuse through matter in nuclear weapons design — a problem with no closed-form solution.

The fundamental idea is remarkably elegant: if you cannot compute a quantity analytically (by formula), *simulate* the underlying random experiment a very large number of times and average the

1.1 What Are Monte Carlo Methods? II

results. Two classical theorems from probability theory guarantee that this procedure is both correct and quantifiably accurate:

- The **Law of Large Numbers** (LLN) guarantees that the average of many independent repetitions converges to the true expected value as the number of repetitions grows without bound — the method is correct in the limit.
- The **Central Limit Theorem** (CLT) tells us precisely how large the error is for a given number of repetitions, allowing us to build confidence intervals around our estimate and know, with a specified probability, how far from the truth our answer is.

1.1 What Are Monte Carlo Methods? III

Why does randomness help? There are three broad classes of problems where deterministic methods struggle and Monte Carlo excels. Understanding these categories helps you decide when Monte Carlo is the right tool.

1. **High-dimensional integration.** Evaluating an integral over a d-dimensional domain (where d is the number of dimensions, also called the "dimension of the problem") using a regular grid requires m^d function evaluations, where m (integer m) is the number of grid points per dimension. This explosive growth is called the *curse of dimensionality*: the cost doubles (or more) every time you add one more dimension. For example, with $m = 10$ grid points per dimension and $d = 20$ dimensions, a regular grid requires 10^{20} evaluations — far beyond any computer that could ever be built. The Monte Carlo estimator, by contrast, achieves a root-mean-squared error of $O(n^{-1/2})$ ("order n to the minus one-half", meaning the error falls proportionally to the reciprocal of the square root of n) regardless of d, where n (integer n) is the number of random samples. This *dimension-independence* is the single most important advantage of Monte Carlo and the reason it dominates all other methods for problems in high dimensions.

2. **NP-hard combinatorial problems.** Problems like the Travelling Salesman Problem (TSP) have a search space containing $n!$ (read "n factorial", meaning $1 \times 2 \times 3 \times \cdots \times n$) possible solutions, which grows faster than any fixed exponential. For $n = 50$ cities, $50! \approx 3 \times 10^{64}$ — exhaustive search is not merely impractical; it is physically impossible. Monte Carlo heuristics such as simulated annealing and the cross-entropy method find near-optimal solutions efficiently by intelligently and randomly exploring the search space.
3. **Complex stochastic systems.** When the dynamics of a system involve intrinsic randomness (for example, queuing networks, financial models, or particle physics), direct simulation is often the only tractable approach. We simply run the system many times and record statistics of interest, such as average waiting times or probabilities of rare events.

Structure of this course. The lectures are structured to build from foundational theory to advanced methods, always connecting mathematical theory to Python implementations. Chapter 1 provides three motivating examples that recur throughout the text: a random walk, the Travelling Salesman Problem, and numerical integration.

1. **Ch. 1** Introduction: three motivating examples — random walks (the arcsine law and the law of the iterated logarithm), the Travelling Salesman Problem, and numerical integration. Comparison of standard Monte Carlo with quasi-Monte Carlo.
2. **Ch. 2** Theory of pseudorandom number generators (PRNGs) and statistical tests for randomness quality.
3. **Ch. 3** Generating random variables from arbitrary distributions: inversion, rejection, and transformation methods.
4. **Ch. 4** Output analysis: confidence intervals, bias, variance estimation, and consistent estimators.
5. **Ch. 5** Variance reduction techniques: stratification, antithetic variables, importance sampling, and the cross-entropy method.

1.1 What Are Monte Carlo Methods? VI

6. **Ch. 6** Markov Chain Monte Carlo (MCMC): the Metropolis–Hastings algorithm, the Gibbs sampler, and coupling from the past (CFTP) for exact sampling.
7. **Ch. 7** Stochastic optimisation: simulated annealing and the cross-entropy method applied to the TSP and other problems.
8. **Ch. 8** Simulation of queues and related stochastic processes.

Key Distinction: MC vs. Quasi-MC

Monte Carlo (MC) uses *pseudo-random* sequences — deterministic sequences that are statistically indistinguishable from true randomness and pass all standard tests for randomness. **Quasi-Monte Carlo (QMC)** uses carefully designed *low-discrepancy* sequences that fill the d-dimensional unit hypercube $[0, 1]^d$ more uniformly than random points, achieving smaller integration errors for smooth functions. QMC can converge faster for smooth integrands in moderate dimensions, but has important limitations for simulation tasks. Sections 1.4 and 1.5 compare the two approaches in depth with concrete numerical examples.

1.1 Pseudorandom Numbers in Python

Before tackling the theory, it is essential to know how to generate pseudo-random numbers in Python. The `NumPy` library provides a modern, reproducible interface via `default_rng()`. Passing a *seed* (any integer you choose as the starting value of the sequence) makes the entire output sequence deterministic and reproducible: if you share your seed with a colleague, they can reproduce your simulation results exactly, step by step.

NumPy Recommended API (NumPy $\geq$ 1.17)

The function `default_rng(seed)` returns a generator object backed by the PCG64 algorithm. Here seed is any non-negative integer you choose; different seeds give different but equally valid streams. Calling `rng.random(k)` draws k (integer k) samples independently and uniformly from $\mathcal{U}[0, 1)$ (the uniform distribution on the interval from 0 inclusive to 1 exclusive). Calling `rng.integers(low=0, high=M, size=k)` draws k integers uniformly from the set $\{0, 1, \ldots, M-1\}$.

```
import numpy as np
from numpy.random import default_rng, PCG64

# Recommended: default_rng with explicit seed for reproducibility
rng = default_rng(seed=31415)              # PCG64 generator, seeded

u   = rng.random(size=5)                   # 5 samples from Uniform[0,1)
b   = rng.integers(low=0, high=2, size=10)  # 10 values in {0,1}
```

1.2.1 Setup: The Coin-Toss Game I

The *simple symmetric random walk* is one of the most fundamental and well-studied objects in probability theory. It models phenomena as diverse as stock price fluctuations, the conformation of polymer chains in solution, the diffusion of particles in a liquid, and the fortune of a gambler. It provides a perfect test bed for Monte Carlo simulation: the exact analytical answers are known from classical limit theorems (due to Feller, 1950), so we can directly *verify* that our simulations converge to the right values. This is precisely how practitioners "validate" simulation code before applying it to problems with unknown answers.

The game (Feller, 1950). Two players, A and B, repeatedly bet on fair coin tosses. Player A wins $+1$ monetary unit from B when the coin lands heads, and loses 1 unit when it lands tails. Player A's cumulative fortune (net winnings) after n (integer n, pronounced "n") tosses is denoted S_n (S subscript n, read "S sub n"). The starting fortune is $S_0 = 0$ (zero before any tosses).

1.2.1 Setup: The Coin-Toss Game II

Definition — Simple Symmetric Random Walk (SSRW)

Let $X_1, X_2, \ldots$ (X subscript 1, X subscript 2, etc., each pronounced "X sub j" for the j-th toss) be independent and identically distributed (i.i.d.) random variables satisfying:

$$\mathbb{P}(X_j = +1) \;=\; \mathbb{P}(X_j = -1) \;=\; \tfrac{1}{2}.$$

Here $\mathbb{P}$ (blackboard bold P, denoting "probability") is the probability of the event in parentheses. The value $X_j = +1$ represents heads on the j-th toss, and $X_j = -1$ represents tails. The index j runs over $1, 2, 3, \ldots$ The **partial sums** are defined by:

$$S_0 = 0, \qquad S_n = \sum_{j=1}^{n} X_j, \quad n = 1, 2, \ldots$$

The symbol $\sum_{j=1}^{n}$ (Sigma, meaning "sum from j equals 1 to n") adds up all the individual steps $X_1 + X_2 + \cdots + X_n$. The sequence $S_0, S_1, S_2, \ldots$ is called the **simple symmetric random walk**.

1.2.1 Setup: The Coin-Toss Game III

How to Read This Formula

S_n represents player A's running total after exactly n coin tosses. Because each step X_j is $+1$ or -1 with equal probability and is independent of all other tosses, the walk is equally likely to go up or down at each step. Note that S_n must lie in the set $\{-n, -n+2, \ldots, n-2, n\}$: since each step is ± 1, the sum of n steps has the same parity (odd or even) as n, so odd sums are impossible when n is even. In particular, $|S_n| \leq n$ always: the walk can never exceed the maximum possible fortune of winning every toss.

Four questions about the walk. We consider a walk of $2N$ steps (an even number, which ensures that ties — both players having equal fortune — can only occur at even times). The following four questions all have surprising and counter-intuitive answers:

Q1 (Last tie time). Let L_{2N} (L subscript 2N) be the *last time up to step $2N$ at which the walk equals zero*, i.e., the last time the two players are exactly equal. Formally, $L_{2N} = \max\{i \leq 2N : S_i = 0\}$. Intuition says this should typically be near the end of the game — is it?

Q2 (Fraction of time winning). Let $P(\alpha, \beta)$ (where α (alpha) and β (beta) are fractions satisfying $0 \leq \alpha < \beta \leq 1$) be the probability that player A is winning for a fraction of time between α and β. Since the game is fair, do you expect this distribution to be concentrated near $1/2$ (equal sharing of the lead)?

Q3 (Size of fluctuations). How large does $|S_n|$ (the absolute value of S_n, measuring how far from zero the walk is) grow with n? Does it grow proportionally to n, to $\sqrt{n}$ (square root of n), or at some other rate?

Q4 (Probability of no tie). What is the probability that one player is *always* winning, i.e., that $S_k > 0$ for all steps $k = 1, \ldots, 2N$? Does this probability tend to zero as N grows?

1.2.1 Setup: The Coin-Toss Game V

All four questions have exact analytical answers from classical limit theorems. Monte Carlo simulation lets us *verify* these answers numerically and develop intuition before studying the mathematical proofs.

1.2.1 Setup: The Coin-Toss Game VI

Formal definitions of the key quantities.

Last Tie Time (L_{2n}) and Time Spent Above Zero (L_{2n}^+)

The *last tie time* is:

$$L_{2n} = \sup\{m \leq 2n : S_m = 0\},$$

where "sup" (supremum, read "soop") means the *largest* element of the set — here, the latest step at which the walk visits zero. If the walk never returns to zero after time 0, then $L_{2n} = 0$.

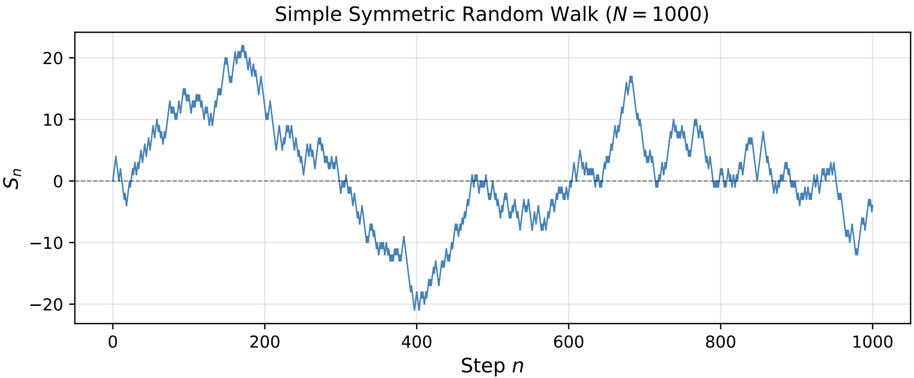
The *time spent above zero* is:

$$L_{2n}^+ = \#\{j = 1, \ldots, 2n : S_{j-1} \geq 0 \text{ or } S_j \geq 0\},$$

which counts the number of steps during which player A is winning (or at least not losing). Here $\#$ (hash, meaning "the number of elements in the set") counts how many steps j satisfy the condition. A step j is counted as "above zero" if *at least one* of the two walk values at its endpoints — S_{j-1} or S_j — is non-negative. The symbol $\geq$ means "greater than or equal to".

1.2.1 Setup: The Coin-Toss Game VII

Both L_{2n} and L_{2n}^{+} grow to infinity as $n \to \infty$ (as the number of steps n grows without bound). However, the *rescaled* fractions $L_{2n}/(2n)$ and $L_{2n}^{+}/(2n)$ both lie in $[0, 1]$, and their distributions converge as $n \to \infty$ to the famous *arcsine distribution* (discussed in detail in Section 1.2.3 below).



A single realisation of the SSRW for $N = 1000$ steps. The walk exhibits long "excursions" (extended periods) on one side of zero — a direct consequence of the arcsine law discussed in Section 1.2.3.

1.2.2 How Large Are the Fluctuations? The Law of the Iterated Logarithm I

Our first quantitative question is: how large does $|S_n|$ (the absolute value of the walk position, measuring how far from zero the player A's fortune has gone) grow as n (the number of steps) increases? To answer this, we try normalising the walk by different functions of n and ask what the normalised sequence converges to.

Three candidate normalisation rates. Since $|S_n| \leq n$ always, we can normalise by any function of n that is between 1 and n. There are three natural candidates, each giving a different type of answer.

- **Normalise by n:** The Strong Law of Large Numbers (SLLN) gives $S_n/n \to 0$ almost surely (with probability 1) as $n \to \infty$. This is because the mean (expected value) of each step X_j is zero: $\mathbb{E}[X_j] = (+1)(1/2) + (-1)(1/2) = 0$. So dividing by n is *too aggressive* — the normalised walk collapses to zero and we learn nothing about the actual fluctuations.
- **Normalise by $\sqrt{n}$ (square root of n):** The Central Limit Theorem (CLT) says $S_n/\sqrt{n}$ converges *in distribution* to $\mathcal{N}(0,1)$ (the standard normal distribution, also called the bell curve). Here $\mathcal{N}(0,1)$ (N zero one) has mean 0 and variance 1. However, the sequence $S_n/\sqrt{n}$ does *not* converge almost surely — it oscillates indefinitely between large positive and large negative values. So $\sqrt{n}$ is *too weak*: the normalised walk does not settle.

1.2.2 How Large Are the Fluctuations? The Law of the Iterated Logarithm II

- **Normalise by $\sqrt{2n \log \log n}$:** This is the *correct* scale, as stated by the Law of the Iterated Logarithm. Here "$\log$" (log) denotes the natural logarithm (base e, where $e \approx 2.71828$), and "$\log \log n$" means the logarithm of the logarithm of n — an extremely slowly growing function of n. The factor "iterated logarithm" in the name refers to this double application of the logarithm.

Law of the Iterated Logarithm (LIL) — Khintchine, 1924; Kolmogorov, 1929

For the simple symmetric random walk $S_n = \sum_{j=1}^{n} X_j$, where each $X_j \in \{-1, +1\}$ with probability $1/2$ each, the following hold with probability 1:

$$\mathbb{P}\left(\limsup_{n\to\infty} \frac{S_n}{\sqrt{2n \log \log n}} = +1\right) = 1 \quad \text{and} \quad \mathbb{P}\left(\liminf_{n\to\infty} \frac{S_n}{\sqrt{2n \log \log n}} = -1\right) = 1.$$

Here "$\limsup_{n\to\infty}$" (limit superior, read "lim-sup as n goes to infinity") is the largest value that the sequence gets close to infinitely often as n grows — intuitively, the highest "ceiling" the sequence keeps touching without permanently exceeding. Symmetrically, "$\liminf_{n\to\infty}$" (limit inferior, or "lim-inf") is

1.2.2 How Large Are the Fluctuations? The Law of the Iterated Logarithm III

the smallest such accumulation point, the lowest floor the sequence keeps touching. The two formulas together say: with probability 1, the walk reaches arbitrarily close to $+\sqrt{2n \log \log n}$ infinitely often, and reaches arbitrarily close to $-\sqrt{2n \log \log n}$ infinitely often, but never permanently exceeds either boundary.

How to Read the LIL

In plain language: the walk S_n never settles down — it oscillates without bound. But it stays within a well-defined almost-sure *envelope*: $|S_n| \leq \sqrt{2n \log \log n}$ for all sufficiently large n, with probability 1. The walk touches the upper boundary $+\sqrt{2n \log \log n}$ infinitely often, and likewise touches the lower boundary $-\sqrt{2n \log \log n}$ infinitely often. The "iterated logarithm" $\log \log n$ is a tiny correction factor: it grows incredibly slowly (for $n = 10^{100}$, we have $\log \log n \approx 5.4$), so the envelope is only slightly wider than the naive $\sqrt{n}$ envelope. This precise scaling is not merely a theoretical curiosity — it has practical consequences for how long you need to run a simulation before results stabilise.

1.2.2 How Large Are the Fluctuations? The Law of the Iterated Logarithm IV

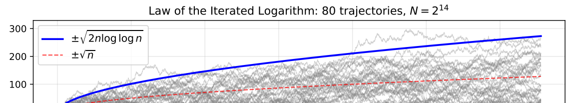
Concrete numerical illustration of $\sqrt{2n \log \log n}$.
Let us plug in specific values to get a feel for how this envelope grows. For $n = 10{,}000$ (ten thousand steps):

1.2.2 How Large Are the Fluctuations? The Law of the Iterated Logarithm V

Worked Numerical Calculation at $n = 10{,}000$

$$\begin{aligned}
\sqrt{n} &= \sqrt{10{,}000} = 100, \\
\log n &= \log(10{,}000) = \log(10^4) = 4\log(10) \approx 4 \times 2.303 \approx 9.21 \quad \text{(natural log)}, \\
\log\log n &= \log(9.21) \approx 2.22, \\
\sqrt{2n\log\log n} &= \sqrt{2 \times 10{,}000 \times 2.22} = \sqrt{44{,}400} \approx 211.
\end{aligned}$$

So the LIL envelope at step $n = 10{,}000$ is approximately ± 211, which is about 2.11 times larger than the "naive" square-root boundary of $\pm\sqrt{n} = \pm 100$. The gap between the two envelopes widens only very slowly (logarithmically) as n grows further.



Simulation evidence. The figure shows 80 independent SSRW trajectories of length $n = 2^{14} = 16{,}384$ steps, each started at $S_0 = 0$. The solid blue curves are the LIL

1.2.2 How Large Are the Fluctuations? The Law of the Iterated Logarithm VI

Key Takeaway: The Correct Scale of Fluctuations

The walk grows neither as slowly as $\sqrt{n}$ nor as fast as n. The precise almost-sure scale is $\sqrt{2n \log \log n}$, which is only slightly (logarithmically) larger than $\sqrt{n}$. This precise scaling has important practical consequences: it tells you how far from the mean a simulation output can stray if you run it for n steps, which informs decisions about how long to run a simulation and how to interpret unexpected extreme outputs.

1.2.3 The Arcsine Law I

The arcsine law provides exact answers to questions Q1 and Q2, and stands as one of the most counter-intuitive results in elementary probability theory. Understanding it thoroughly will permanently change how you think about "fairness" in random processes.

What intuition suggests — and why it is wrong. Since the walk is perfectly symmetric (the probability of going up equals the probability of going down at every step), you might expect the fraction of time that player A is winning to be typically close to $1/2$. After all, the game is fair: shouldn't both players lead for roughly equal portions of the time? The arcsine law gives the precise, surprising answer: being ahead for *nearly all* of the time or *nearly none* of the time is *most likely*, while being ahead for exactly half the time is the *least likely* outcome of all.

Arcsine Law (Feller, 1950; Durrett, 1996)

For $0 \leq \alpha < \beta \leq 1$ (where α (alpha) and β (beta) are fractions between 0 and 1 representing lower and upper bounds on the fraction of time player A leads), as $n \to \infty$ (as the number of steps grows without bound), the fraction of time spent above zero converges in distribution:

$$\mathbb{P}\left(\alpha \leq \frac{L_{2n}^{+}}{2n} \leq \beta\right) \longrightarrow \int_{\alpha}^{\beta} \frac{dx}{\pi\sqrt{x(1-x)}}. \tag{1.2.1}$$

1.2.3 The Arcsine Law III

Notation for the Arcsine Law Formula

In this formula:

- $\mathbb{P}(\cdot)$ (blackboard P) is the probability of the event described.
- L_{2n}^{+} (L-plus subscript 2n) counts the number of steps during which player A is winning.
- $L_{2n}^{+}/(2n)$ is the total *fraction* of the total $2n$ steps during which player A leads, a number between 0 and 1.
- α (alpha) and β (beta) are two fractions with $0 \leq \alpha < \beta \leq 1$.
- $\int_{\alpha}^{\beta}$ (integral from α to β) computes the area under the curve $1/(\pi\sqrt{x(1-x)})$ between $x = \alpha$ and $x = \beta$.
- π (pi) is the mathematical constant ≈ 3.14159.
- $x(1-x)$ (x times one minus x) is zero at both endpoints $x = 0$ and $x = 1$, so the density $1/(\pi\sqrt{x(1-x)})$ is infinitely large there.

1.2.3 The Arcsine Law IV

Why is it called the "arcsine" law? The cumulative distribution function of the limiting distribution (the probability that the fraction of winning time is at most β) can be computed in closed form by evaluating the integral:

$$\int_0^\beta \frac{dx}{\pi\sqrt{x(1-x)}} = \frac{2}{\pi}\arcsin\big(\sqrt{\beta}\big),$$

where $\arcsin$ (arcsine, the inverse of the sine function) appears explicitly. This closed form is where the distribution gets its name.
The corresponding probability density function (PDF) is:

$$p(x) = \frac{1}{\pi\sqrt{x(1-x)}}, \qquad 0 < x < 1.$$

This density has a characteristic **U-shape**: it equals $+\infty$ at $x = 0$ and at $x = 1$ (so the walk is most likely to spend all its time on one side of zero, or none of its time there), and achieves its minimum value of $2/\pi \approx 0.637$ at $x = 1/2$ (the least likely outcome is a perfectly equal split of the lead). The

1.2.3 The Arcsine Law V

U-shape is the mathematical signature of the gambler's fallacy: extreme outcomes are more common than moderate ones.

Worked Numerical Calculation: Probability of a "Fair" vs. "Extreme" Outcome

The probability that player A leads between 40% and 60% of the time (a "near-equal" split):

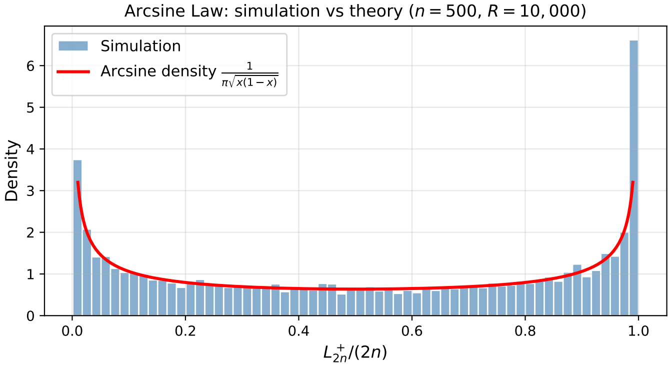
$$\mathbb{P}\left(0.4 \le \frac{L_{2n}^{+}}{2n} \le 0.6\right) \approx \frac{2}{\pi}\arcsin(\sqrt{0.6}) - \frac{2}{\pi}\arcsin(\sqrt{0.4}) \approx 0.795 - 0.667 \approx 0.128.$$

Only about 12.8% of long games end with a near-equal split of the lead!
The probability of an "extreme" outcome (one player leads more than 90% of the time):

$$\mathbb{P}\left(\frac{L_{2n}^{+}}{2n} \le 0.1\right) + \mathbb{P}\left(\frac{L_{2n}^{+}}{2n} \ge 0.9\right) \approx 2 \times \frac{2}{\pi}\arcsin(\sqrt{0.1}) \approx 2 \times 0.205 \approx 0.410.$$

Over 41% of games end with one player dominating for more than 90% of the time — more than three times as common as the near-equal split! This is the arcsine law in action.

1.2.3 The Arcsine Law VI



1.2.3 The Arcsine Law VII

Histogram of $L^{+}_{1000}/1000$ *(fraction of 1000-step walks spent above zero) from* $R = 10{,}000$ *independent replications. The red curve is the theoretical arcsine density* $p(x) = 1/(\pi\sqrt{x(1-x)})$*. The U-shape is clearly visible: the bars at both extremes (near 0 and near 1) are tallest, while the bar near* $x = 1/2$ *is shortest. The simulation matches the theory closely.*

Key Insight: The Gambler's Fallacy Refuted

The arcsine law overturns the naive intuition that "fair" games lead to equal sharing of the lead. A player who is behind for the first 99% of a game is *not* especially likely to "recover" and take the lead before the end. In fact, being behind (or ahead) for the entire game is a completely typical and mathematically expected outcome. This is the *gambler's fallacy* refuted: past losses do not make future wins more likely in a symmetric game, because the walk has no memory. Each coin toss is independent of all previous ones.

Connection to randomness testing. The arcsine law is not merely a theoretical curiosity about gambling. In Section 2.5, we will use the arcsine law to construct a *statistical test* for the quality of a pseudorandom number generator: if the generator truly produces random numbers, then the statistic

1.2.3 The Arcsine Law VIII

$L_{2n}^{+}/(2n)$ computed from its output should follow the arcsine distribution. A significant deviation from the arcsine distribution signals a defect or pattern in the generator.

1.2.4 Python: Simulating the SSRW and the LIL Envelope

The following code simulates a single SSRW trajectory and overlays the LIL envelope $\pm\sqrt{2n \log\log n}$ to visually verify the theorem. The key implementation step is converting uniform integers in $\{0, 1\}$ to steps in $\{-1, +1\}$ via the formula $X_j = 1 - 2B_j$ where $B_j \in \{0, 1\}$: when $B_j = 0$ we get $X_j = 1$ (heads), and when $B_j = 1$ we get $X_j = -1$ (tails). The function `np.cumsum` (short for "cumulative sum") then computes all partial sums $S_1, S_2, \dots, S_{2N}$ at once as an array, without any explicit loop.

```
import numpy as np
import matplotlib.pyplot as plt
from numpy.random import default_rng

rng = default_rng(seed=31415)
N   = 1000                                # walk has 2N steps

steps = 1 - 2 * rng.integers(0, 2, size=2*N)  # {-1, +1}
S     = np.concatenate([[0], steps.cumsum()])  # S_0=0, S_1,...,S_{2N}

# LIL envelope: +/- sqrt(2 * n * log(log(n)))
# We start from n=2 to avoid log(log(1))=log(0) which is -infinity
n_arr    = np.arange(2, 2*N + 1)
envelope = np.sqrt(2 * n_arr * np.log(np.log(n_arr)))

fig, ax = plt.subplots(figsize=(10, 3.5))
ax.plot(range(2*N + 1), S, lw=0.8, color='steelblue', label='$S_n$')
ax.plot(n_arr,  envelope, 'b-', lw=1.8, alpha=0.7,
        label=r'LIL: $\pm\sqrt{2n\log\log n}$')
ax.plot(n_arr, -envelope, 'b-', lw=1.8, alpha=0.7)
```

1.2.5 Python: Verifying the Arcsine Law

To verify the arcsine law statistically, we simulate $R = 10{,}000$ independent walks of $2n = 1000$ steps each, compute $L^+_{2n}/(2n)$ for each walk, and compare the empirical histogram to the theoretical arcsine density. Vectorising the simulation over all R replications simultaneously using NumPy's array operations (operating on the entire matrix at once rather than looping over replications) makes this fast enough to run interactively on a laptop in a few seconds.

```
import numpy as np
import matplotlib.pyplot as plt
from numpy.random import default_rng

def fraction_above_zero(n=500, R=10_000, seed=0):
    """Return array of L^+_{2n}/(2n) for R independent walks.

    n  : half-length (walk has 2n steps)
    R  : number of independent replications
    Returns: array of shape (R,), each value in [0,1]
    """
    rng   = default_rng(seed)
    # shape: (R, 2n) -- each row is one walk's steps
    steps = 1 - 2 * rng.integers(0, 2, size=(R, 2*n))
    # cumsum along axis=1, prepend column of zeros --> shape (R, 2n+1)
    S = np.hstack([np.zeros((R, 1)), steps.cumsum(axis=1)])
    # Step j is counted if S_{j-1}>=0 OR S_j>=0 (at least one endpoint >= 0)
    above = (S[:, :-1] >= 0) | (S[:, 1:] >= 0)
    return above.sum(axis=1) / (2*n)   # shape (R,), each in [0,1]
```

1.3.1 The TSP: Problem Statement and Why It Is Hard I

The Travelling Salesman Problem (TSP) is a paradigmatic hard combinatorial optimisation problem. It arises in many real-world contexts: logistics (planning delivery routes for trucks), circuit board manufacturing (minimising the travel distance of a drilling machine), genome sequencing (ordering DNA fragments along a chromosome), and telescope scheduling (minimising slew time between observations). It is our second motivating example for Monte Carlo methods, because it illustrates why randomised heuristics are indispensable when exact search is computationally infeasible.

The problem. A salesman must visit n cities exactly once and return to the starting city, forming a closed tour. The distance (or cost) of travelling from city i to city j is given by the entry $\mathbf{M}_{ij}$ (bold M subscript ij, read "M sub i j", a real number representing the travel cost) in an $n \times n$ matrix $\mathbf{M}$ (bold M, an n-by-n array of all pairwise distances). The entry is $\mathbf{M}_{ij} = \infty$ (infinity) if there is no direct route from city i to city j. The goal is to find the *shortest closed tour* — a route that starts and ends at the same city and visits each city exactly once, with the smallest total travel distance.

Mathematically, a *tour* is a permutation σ (sigma (sig-ma), a re-ordering or rearrangement) of the cities $\{1, \ldots, n\}$. The expression $\sigma(k)$ (sigma of k) denotes the k-th city visited. The total length (total distance) of the tour is:

$$l(\sigma) = \sum_{k=1}^{n-1} \mathbf{M}_{\sigma(k),\, \sigma(k+1)} + \mathbf{M}_{\sigma(n),\, \sigma(1)}.$$

1.3.1 The TSP: Problem Statement and Why It Is Hard II

The sum $\sum_{k=1}^{n-1}$ (sigma from k equals 1 to $n-1$) adds up the distances of all consecutive legs of the tour except the final return leg; the last term $\mathbf{M}_{\sigma(n),\sigma(1)}$ adds the distance from the last city back to the first, closing the loop.

Optimisation Problem: Find the Shortest Tour

We seek the optimal permutation (the ordering of cities that gives the shortest total distance):

$$\sigma^* = \arg\min_{\sigma \in S_n} l(\sigma),$$

where S_n (S subscript n) denotes the set of all $n!$ permutations of the n cities, and "$\arg\min$" (argument of the minimum) means "the permutation σ for which $l(\sigma)$ achieves the minimum value." The symbol σ^* (sigma-star) conventionally denotes the optimal solution.

1.3.1 The TSP: Problem Statement and Why It Is Hard III

Why is the TSP computationally hard? The search space S_n contains $n!$ (n factorial, meaning $1 \times 2 \times 3 \times \cdots \times n$) possible tours. The factorial function grows faster than any fixed exponential, as the following table dramatically illustrates:

n cities	$n!$ permutations	Time to check all at 10^9/sec
10	$3{,}628{,}800 \approx 3.6 \times 10^6$	0.004 seconds
15	$\approx 1.3 \times 10^{12}$	21 minutes
20	$\approx 2.4 \times 10^{18}$	76 years
50	$\approx 3.0 \times 10^{64}$	longer than the age of the universe

At 10^9 (one billion) tour evaluations per second, evaluating all $50!$ tours would require approximately 10^{55} years. The current age of the universe is only about 1.4×10^{10} years. Exhaustive search is not just slow — it is completely and permanently infeasible for $n \geq 25$ or so, regardless of how fast computers become.

The TSP belongs to the class of **NP-complete** problems: no algorithm is known that solves all instances in polynomial time (time proportional to some fixed power n^k), and the overwhelming

1.3.1 The TSP: Problem Statement and Why It Is Hard IV

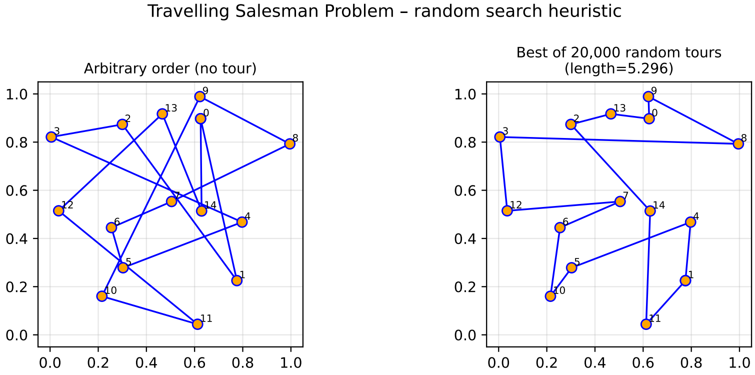
consensus among theoretical computer scientists is that no such algorithm exists. Exact algorithms using clever branch-and-bound methods are practical only for $n \lesssim 50$ (and can take hours even then).

Monte Carlo heuristics for the TSP. Since exact optimisation is infeasible for large n, we use *randomised approximation algorithms* that efficiently find near-optimal (but not necessarily exactly optimal) solutions.

Random restart. Sample R (integer R, the number of trials) random permutations uniformly from S_n, evaluate $l(\sigma)$ for each, and return the best one found. This is the simplest Monte Carlo approach, analogous to drawing lottery tickets and keeping the best. For $n = 15$ cities and $R = 20{,}000$ trials, random search typically finds a tour that is 30–50% longer than the optimal tour.

Simulated Annealing (Ch. 7). Start with a random tour. At each step, propose a small local modification called a *2-opt move*: remove two edges from the tour and reconnect the four endpoints differently, reversing the segment between them. Always accept improvements. Accept worsening moves with probability $\exp(-\Delta l/T)$ (read "exponential of minus delta-l over T"), where Δl (delta l, the Greek letter delta meaning "change in") is the increase in tour length due to the move, and T (T) is a "temperature" parameter that decreases slowly over time. Accepting worsening moves with some probability allows escape from local optima. This is the key insight behind simulated annealing.

Cross-Entropy Method (Chs. 5, 7). Maintain a probability distribution over tours, sample batches of tours from it, keep the best fraction (the "elite sample"), and update the distribution to assign higher probability to elite tours. Repeating this cycle drives the distribution toward near-optimal tours.



Left: 15 random cities connected in an arbitrary random order (no optimisation; many edge crossings visible, which is always suboptimal). Right: best tour found by random search over $R = 20{,}000$ *random permutations. The absence of crossings is a necessary (but not sufficient) condition for an optimal tour.*

1.3.2 Python: Random Tour Search for the TSP

The following implementation evaluates $R = 20{,}000$ random tours of $n = 15$ cities placed uniformly at random in the unit square $[0, 1]^2$, and keeps the shortest tour found. The function `tour_length` uses `np.roll` (shift all elements of the array by one position, wrapping the last element to the front) to implement the cyclic distance computation efficiently without an explicit loop over cities.

```python
import numpy as np
from numpy.random import default_rng

def tour_length(cities, perm):
    """Compute the total Euclidean length of a closed tour.

    cities : shape (n, 2) -- the (x, y) coordinates of each city
    perm   : permutation of {0, ..., n-1} -- the order to visit cities
    Returns: the total distance of the tour (a single float)
    """
    # shifted[k] = perm[k+1 mod n], so it gives the NEXT city after perm[k]
    # This creates the "closing the loop" effect automatically
    shifted = np.roll(perm, -1)
    # Compute (x, y) differences for each leg of the tour: shape (n, 2)
    diffs   = cities[perm] - cities[shifted]
    # Euclidean distance of each leg: sqrt(dx^2 + dy^2), then sum all legs
    return np.linalg.norm(diffs, axis=1).sum()

def random_tour_search(cities, R=20_000, seed=42):
    """Return (best_tour, best_length) from R random permutations.
```

1.4.1 The Monte Carlo Integration Framework I

One of the most important and widely used applications of Monte Carlo methods is numerical estimation of integrals, particularly in high dimensions where classical deterministic quadrature methods become hopelessly expensive. This section develops the basic theory and shows why MC is *completely dimension-independent* in its convergence rate — a remarkable property that no grid-based quadrature method can match.

The integration problem. We want to compute the integral:

$$I = \int_{[0,1]^d} k(\mathbf{u})\, d\mathbf{u},$$

1.4.1 The Monte Carlo Integration Framework II

Notation for the Integration Problem

In this formula:

- $k : [0, 1]^d \to \mathbb{R}$ (k, a function) maps each point in the d-dimensional unit hypercube to a real number. This is the function whose integral we want.
- $[0, 1]^d$ (the d-dimensional unit hypercube) is the set of all d-dimensional vectors $\mathbf{u} = (u_1, \ldots, u_d)$ where each coordinate u_i lies between 0 and 1.
- d (the dimension) is the number of coordinates, which can be very large (e.g., $d = 100$ or $d = 1000$).
- $\mathbf{u}$ (bold u) is the variable of integration, a d-dimensional vector.
- $d\mathbf{u}$ denotes integration with respect to all d coordinates simultaneously.

This formulation covers a large class of problems: any integral over a bounded domain can be transformed to this form by a change of variables.

1.4.1 The Monte Carlo Integration Framework III

The probabilistic interpretation. If $\mathbf{U}$ (bold U, a random vector) is uniformly distributed on $[0,1]^d$ (meaning each coordinate is independently and uniformly distributed on $[0,1)$), then the integral equals the expected value:

$$I \;=\; \mathbb{E}[k(\mathbf{U})].$$

This identity — the integral equals the expected value of k at a uniform random point — is the fundamental bridge between integration and probability that makes Monte Carlo work. It tells us: to estimate an integral, draw random points uniformly and average the function values.

1.4.1 The Monte Carlo Integration Framework IV

Monte Carlo Estimator

Let $\mathbf{U}_1, \mathbf{U}_2, \ldots, \mathbf{U}_n$ be i.i.d. (independent and identically distributed) uniform random vectors on $[0,1]^d$, where n is the number of samples. Define the **Monte Carlo estimator**:

$$\hat{Y}_n = \frac{1}{n} \sum_{i=1}^{n} k(\mathbf{U}_i). \tag{1.2.2}$$

Here $\hat{Y}_n$ (Y-hat subscript n; the hat $\hat{}$ means "estimate" or "approximation") is the sample average of n independent evaluations of k. The index i runs from 1 to n, and the sum $\sum_{i=1}^{n}$ (Sigma from i equals 1 to n) adds all n values. By the Strong Law of Large Numbers (SLLN), $\hat{Y}_n \to I$ almost surely (with probability 1) as $n \to \infty$. The estimator is **unbiased**: $\mathbb{E}[\hat{Y}_n] = I$ for every n, meaning it is correct on average for any number of samples.

Variance and Error Rate of the MC Estimator

Since the $\mathbf{U}_i$ are i.i.d., the variance (the squared spread around the mean) of $\hat{Y}_n$ is:

$$\mathrm{Var}(\hat{Y}_n) = \frac{\sigma^2}{n}, \quad \text{where} \quad \sigma^2 = \mathrm{Var}(k(\mathbf{U})) = \int_{[0,1]^d} k(\mathbf{u})^2 \, d\mathbf{u} - I^2.$$

Here σ^2 (sigma-squared, where σ (sigma) is the Greek letter for "s") is the variance of a single evaluation $k(\mathbf{U})$, measuring how spread out single evaluations are around the true integral I. The quantity σ (sigma) is the standard deviation: it measures the typical deviation of one evaluation from the mean, in the same units as k. By the Central Limit Theorem (CLT), for large n:

$$\sqrt{n}\,(\hat{Y}_n - I) \xrightarrow{d} \mathcal{N}(0, \sigma^2),$$

meaning $\sqrt{n}$ (square root of n) times the error converges *in distribution* (symbolised by $\xrightarrow{d}$) to a normal (bell-curve) distribution with mean 0 and variance σ^2. The root-mean-squared error (RMSE) of the estimator is:

$$\mathrm{RMSE}(\hat{Y}_n) = \sqrt{\mathbb{E}\big[(\hat{Y}_n - I)^2\big]} = \frac{\sigma}{\sqrt{n}} = O(n^{-1/2}).$$

1.4.2 Python: Estimating π with Monte Carlo

The classic demonstration of MC integration is estimating $\pi \approx 3.14159$ (pi, the ratio of a circle's circumference to its diameter). A quarter of the unit disk — the set $\{(x, y) : x^2 + y^2 \leq 1,\ x \geq 0,\ y \geq 0\}$ — has area $\pi/4$. By throwing random points uniformly in the unit square $[0, 1]^2$ and counting how many land inside the quarter disk, we estimate the area $\pi/4$, and hence π itself.

Formally, we define $k(u_1, u_2) = 4 \cdot \mathbf{1}[u_1^2 + u_2^2 \leq 1]$ (four times the indicator function: $\mathbf{1}[\cdot]$ equals 1 if the condition in brackets is true and 0 otherwise), so that $I = \mathbb{E}[k(\mathbf{U})] = 4 \cdot (\pi/4) = \pi$. The factor of 4 converts the area of the quarter disk to π.

```
import numpy as np
from numpy.random import default_rng

def estimate_pi(n, seed=0):
    """Estimate pi by the quarter-circle Monte Carlo method.

    n    : number of random points to use
    seed : random seed for reproducibility
    Returns: estimate of pi (a float near 3.14159)
    """
    rng    = default_rng(seed)
    U      = rng.random((n, 2))              # n points, each (x,y) in [0,1)^2
    inside = (U**2).sum(axis=1) <= 1.0       # True if x^2 + y^2 <= 1
    return 4.0 * inside.mean()              # 4 * (fraction inside)
```

1.5.1 The Idea Behind Quasi-Monte Carlo I

The MC error rate $O(n^{-1/2})$ arises directly from the *randomness* of the sample points: random points inevitably form clusters and leave empty regions, resulting in discrepancy $D = O(n^{-1/2})$. This raises a natural and important question: can we do better by replacing random points with *carefully constructed deterministic* points that fill $[0, 1]^d$ more uniformly than random points can?

The answer is yes, and this is the central idea of **Quasi-Monte Carlo (QMC)** methods. Instead of using genuinely (or pseudo-)random points, QMC uses *low-discrepancy sequences* — deterministic sequences designed to spread points as evenly as possible throughout the space.

1.5.1 The Idea Behind Quasi-Monte Carlo II

Low-Discrepancy Sequences

A sequence $\mathbf{x}_1, \mathbf{x}_2, \ldots \in [0,1)^d$ is called a *low-discrepancy sequence* (or *quasirandom sequence*) if its star-discrepancy satisfies:

$$D(\mathbf{x}_1, \ldots, \mathbf{x}_n; \mathcal{B}^*) = O\left(\frac{(\log n)^d}{n}\right).$$

Here $(\log n)^d$ (log-n raised to the power d, the dimension) is a dimension-dependent logarithmic factor; for fixed d, this expression grows much more slowly than $\sqrt{n}$ as n increases. Consequently, for fixed d, the discrepancy $O((\log n)^d/n)$ is much smaller than the MC discrepancy $O(n^{-1/2})$ for large n.

For a low-discrepancy sequence, the QMC estimator $\hat{I}_n^{\mathrm{QMC}} = \frac{1}{n}\sum_{i=1}^n k(\mathbf{x}_i)$ converges to I at rate $O(n^{-(1-\varepsilon)})$ (order n to the power minus one, with a tiny correction ε (epsilon) representing any small positive number) for any $\varepsilon > 0$, under mild smoothness conditions on k. For smooth integrands in moderate dimensions, this is dramatically faster than the $O(n^{-1/2})$ MC rate.

1.5.1 The Idea Behind Quasi-Monte Carlo III

Koksma–Hlawka Inequality (Deterministic Error Bound)

If $V_{\mathrm{HK}}(k)$ (V-HK of k, the Hardy–Krause variation of k, named after mathematicians Hardy and Krause) measures the total variation of k over $[0,1]^d$ — roughly, how much the function oscillates — then the QMC error satisfies the *deterministic* bound:

$$\left|\hat{I}_n^{\mathrm{QMC}} - I\right| \le V_{\mathrm{HK}}(k) \cdot D(\mathbf{x}_1, \ldots, \mathbf{x}_n).$$

This inequality, due to Koksma (1942) and Hlawka (1961), says the QMC error is bounded by the product of two factors: one measuring how "bumpy" the function is ($V_{\mathrm{HK}}(k)$), and one measuring how uniformly the points cover the space (D). If both are small, the error is small.

This is a *deterministic* (non-random, worst-case) error bound — it holds for every integrand with finite Hardy–Krause variation, not just on average. No such deterministic bound exists for random MC; for MC we can only give *probabilistic* guarantees (the error is small with high probability).

1.5.1 The Idea Behind Quasi-Monte Carlo IV

The best known low-discrepancy sequences (Halton, Sobol', Faure) satisfy:

$$D(\mathbf{x}_1, \ldots, \mathbf{x}_n) \;\leq\; C_d \cdot \frac{(\log n)^d}{n} + O\left(\frac{(\log n)^{d-1}}{n}\right),$$

where C_d (C subscript d) is a positive constant that depends on the dimension d but not on n. For small d and large n, this bound is much smaller than the MC discrepancy $O(n^{-1/2})$. However, the constant C_d grows rapidly with d, so for very large dimensions ($d \gtrsim 50$), QMC's advantage over MC shrinks and can disappear.

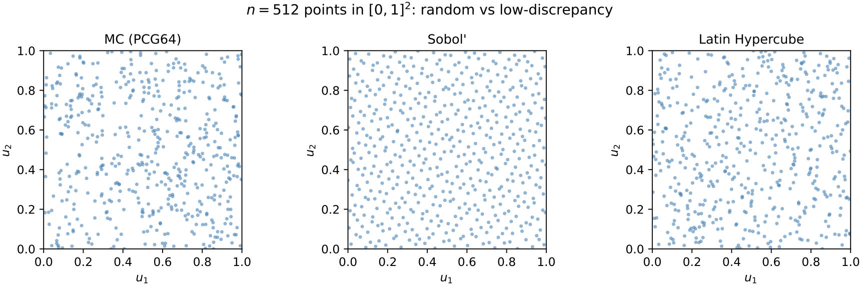
Popular low-discrepancy sequences. All of the following are available in Python via the `scipy.stats.qmc` module (SciPy version 1.7 and later):

Halton sequence. Generalises the Van der Corput sequence to d dimensions by using a different prime number as the base for each dimension. For example, in 2 dimensions: dimension 1 uses base 2 (binary fractions), dimension 2 uses base 3 (ternary fractions). Simple to understand and implement, and works well for small d. For large dimensions, correlations between sequences based on large primes can appear.

Sobol' sequence. Constructed using binary arithmetic and Gray codes (a binary encoding scheme where consecutive integers differ in only one bit). Excellent uniformity for $d \geq 2$. Works best when $n = 2^k$ (a power of two) for some positive integer k. Widely used in financial mathematics for option pricing.

Faure sequence. Based on permuted base-p representations of integers. Has rigorous discrepancy bounds with explicitly computable constants C_d.

Latin Hypercube Sampling (LHS). Divides each coordinate axis into n equally spaced intervals and ensures exactly one sample falls in each interval for each coordinate. This guarantees perfect marginal uniformity (each dimension is perfectly covered) but is not a strictly low-discrepancy sequence in the full d-dimensional sense.



1.5.1 The Idea Behind Quasi-Monte Carlo VII

$n = 512$ points in $[0,1]^2$. Left: MC (PCG64) — visible clusters and empty gaps due to randomness. Centre: Sobol' — noticeably more regular, no obvious gaps or clusters. Right: Latin Hypercube — each row and column of the $\sqrt{n} \times \sqrt{n}$ grid contains exactly one point (perfect marginal uniformity).

1.5.2 The Golden-Ratio and Kronecker Sequences I

Besides the Halton and Sobol' sequences, there is a beautiful and elegant class of low-discrepancy sequences based on *irrational rotations* of the unit interval. These are easy to implement, have a compelling mathematical justification, and exhibit excellent practical performance in one dimension.

1.5.2 The Golden-Ratio and Kronecker Sequences II

The Golden-Ratio (Steinhaus) Sequence in 1D

Define the **golden ratio**:

$$\alpha = \frac{\sqrt{5}-1}{2} \approx 0.6180\ldots$$

Here α (alpha) is the positive root of the quadratic equation $z^2 + z - 1 = 0$; it satisfies $\alpha = 1/(1+\alpha)$, a self-referential identity. The golden-ratio sequence is defined by taking the *fractional parts*:

$$x_n = \{n\alpha\} = n\alpha - \lfloor n\alpha \rfloor, \quad n = 1, 2, 3, \ldots$$

Here $\{x\}$ (braces around x) denotes the *fractional part* of x: for example, $\{3.7\} = 0.7$ and $\{5.2\} = 0.2$ (everything after the decimal point). The symbol $\lfloor x \rfloor$ (floor of x, read "floor x") is the largest integer not exceeding x: for example, $\lfloor 3.7 \rfloor = 3$ and $\lfloor 5.2 \rfloor = 5$. The resulting sequence $(x_n)_{n \geq 1}$ is a low-discrepancy sequence on $[0, 1)$.

1.5.2 The Golden-Ratio and Kronecker Sequences III

Why is the golden ratio special? The golden ratio has the remarkable *continued fraction expansion*:

$$\alpha = \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cdots}}}$$

where all the "partial quotients" (the integers in the denominators) equal 1. In number theory, a real number is *easily approximated by rationals* if it has large partial quotients in its continued fraction: such numbers are "almost rational" and produce sequences that cluster near rational fractions, leaving large gaps. The golden ratio, having all partial quotients equal to 1 (the smallest possible value), is the *most irrational number* — the one that is hardest to approximate by any rational fraction. This makes the golden ratio sequence the most evenly spread one-dimensional sequence achievable by the irrational rotation construction.

1.5.2 The Golden-Ratio and Kronecker Sequences IV

Kronecker Sequences (Generalisation)

More generally, $x_n = \{n\beta\}$ (the fractional part of n times β) is a low-discrepancy sequence whenever β (beta) has a continued fraction expansion with *bounded partial quotients*:

$$\beta = \cfrac{1}{b_1 + \cfrac{1}{b_2 + \cfrac{1}{b_3 + \cdots}}},$$

where each partial quotient b_j (b subscript j) satisfies $b_j \leq M$ for some fixed bound M and for all $j = 1, 2, 3, \ldots$ The golden ratio is the special case $b_j = 1$ for all j, giving the tightest possible bound $M = 1$. These sequences are called *Kronecker sequences* after the mathematician Leopold Kronecker.

Multidimensional Kronecker sequences. The natural extension to d dimensions is the sequence $(\{n\beta_1\}, \ldots, \{n\beta_d\})$ (taking the fractional parts of n times d different irrational numbers, one per dimension, all evaluated at the same integer n). Whether such sequences are provably low-discrepancy in $d \geq 2$ dimensions is an *open problem* in number theory. However, extensive numerical experiments suggest they behave well in practice for moderate d.

Critical Warning: Never Slice a 1D Sequence into d-Vectors

It is *incorrect* to take a 1-dimensional low-discrepancy sequence $x_1, x_2, \ldots$ and group consecutive values into d-dimensional vectors: $(x_1, \ldots, x_d), (x_{d+1}, \ldots, x_{2d}), \ldots$. The resulting d-vectors do *not* inherit the low-discrepancy property in d dimensions — in 2D, such pairs lie along diagonal lines rather than filling the square uniformly, introducing strong systematic correlations. Each dimension must be constructed simultaneously with a separate irrational base, as the `scipy.stats.qmc` library does correctly.

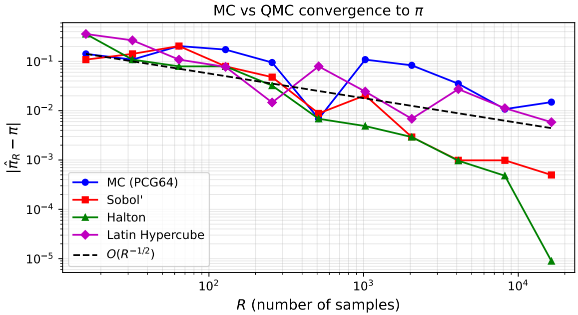
1.5.3 Python: QMC Sequences with `scipy.stats.qmc`

The `scipy.stats.qmc` module (SciPy version 1.7 and later) provides implementations of all major low-discrepancy sequences with a consistent interface. Each class has a `random(n)` method that returns an $n \times d$ NumPy array of quasi-random points in $[0, 1)^d$ (a matrix with n rows and d columns, one point per row). Setting `scramble=True` adds a random linear scrambling that preserves the low-discrepancy property while making the estimator unbiased (so that its expected value equals the true integral, removing any systematic bias from the deterministic structure).

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats.qmc import Sobol, Halton, LatinHypercube
from numpy.random import default_rng

def estimate_pi_from(U):
    """Estimate pi from an n-by-2 array U of points in [0,1)^2."""
    return 4.0 * ((U**2).sum(axis=1) <= 1.0).mean()

# Compare convergence over a range of sample sizes (powers of 2)
# Sobol' works best for n = 2^k, so we use powers of 2 throughout
sample_sizes = [2**k for k in range(4, 15)]  # 16, 32, 64, ..., 16384
mc_err, sob_err, hal_err, lhs_err = [], [], [], []

for n in sample_sizes:
    rng = default_rng(seed=0)
    # MC: pseudorandom points (independent uniform)
    mc_err.append(abs(estimate_pi_from(rng.random((n, 2))) - np.pi))
```

1.6.1 Example 1: Estimating π — Where QMC Wins I

The figure below shows the absolute error $|\hat{\pi}_n - \pi|$ (the vertical axis, on a *logarithmic* scale, meaning each grid line represents a factor of 10) as a function of sample size n (horizontal axis, also logarithmic). On a log-log plot (both axes on logarithmic scale), a convergence rate of $O(n^{-r})$ appears as a straight line with slope $-r$. Therefore, the $O(n^{-1/2})$ MC rate appears as a line with slope $-1/2$, and the $O(n^{-1})$ QMC rate appears as a steeper line with slope -1.



Reading the plot.

- The black dashed line is the reference $O(n^{-1/2})$ slope. The blue MC curve follows this slope closely, confirming the theoretical prediction.
- Sobol' (red) and Halton (green) lie well *below* the MC line at all sample sizes, meaning they achieve much smaller errors for the same n.
- For large n, the QMC slopes are steeper than $-1/2$, approaching -1 for this smooth integrand, as predicted by the Koksma–Hlawka inequality.

1.6.1 Example 1: Estimating π — Where QMC Wins II

Why does QMC work so well for estimating π?

The integrand is $k(u_1, u_2) = 4 \cdot \mathbf{1}[u_1^2 + u_2^2 \leq 1]$ (four times the indicator of the quarter-disk). This function is smooth (constant, in fact) everywhere except along the quarter-circle boundary, where it has a jump discontinuity. The Hardy–Krause variation $V_{\mathrm{HK}}(k)$ is finite (because the boundary is a smooth curve of finite length), so the Koksma–Hlawka inequality applies with QMC discrepancy $D_n = O((\log n)^2/n)$, giving QMC error approaching $O(n^{-1})$.

However, this is a borderline case: for integrands with jump discontinuities *aligned with the coordinate axes* (e.g., $k(u_1, u_2) = \mathbf{1}[u_1 \leq 0.3]$, which is discontinuous along the vertical line $u_1 = 0.3$), the Hardy–Krause variation becomes infinite and QMC may not converge faster than MC.

When QMC Wins and When It Does Not

QMC is effective for *smooth integrands in moderate dimensions* ($d \lesssim 10$–20, where $\lesssim$ means "approximately less than or equal to"). For non-smooth functions (large jumps or kinks), very high dimensions ($d \gtrsim 50$), or — critically — *simulation tasks where the number of random numbers consumed per trial varies* (such as any sequential simulation or Markov chain), QMC can perform *worse* than standard MC. The next section illustrates this failure mode with a concrete example.

1.6.2 Example 2: A 2-D Game — Where QMC Fails I

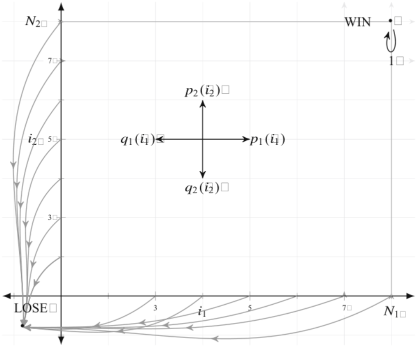
The second comparison example illustrates a setting where quasi-random numbers are not merely less effective but are actively *harmful* — they introduce systematic bias and cause the estimator to converge to the *wrong value*. Understanding why this happens is crucial for anyone using Monte Carlo in practice.

The 2-D game (Lefebvre, L'Ecuyer & Simard). Consider a rectangular grid with $N_1 \times N_2$ states (an N_1-by-N_2 grid of positions). We start at state (i_1, i_2) (position i_1 along the first axis and i_2 along the second axis) and move randomly at each step. The game ends when one of two things happens:

- If any coordinate hits 0 (the bottom or left boundary): we **LOSE**.
- If both coordinates reach their respective maxima ($i_1 = N_1$ and $i_2 = N_2$ simultaneously, the top-right corner): we **WIN**.

From state (i_1, i_2), the four possible transitions at each step are: move right $(i_1 \to i_1 + 1)$ with probability $p_1(i_1)$, move left $(i_1 \to i_1 - 1)$ with probability $q_1(i_1)$, move up $(i_2 \to i_2 + 1)$ with probability $p_2(i_2)$, or move down $(i_2 \to i_2 - 1)$ with probability $q_2(i_2)$, where the remaining probability corresponds to no move.

1.6.2 Example 2: A 2-D Game — Where QMC Fails II



State space of the 2-D game. LOSE is the bottom-left boundary (any coordinate reaches 0); WIN is the top-right corner (both coordinates reach their maxima simultaneously). Arrows show the four possible moves from a generic interior state (i_1, i_2).

1.6.2 Example 2: A 2-D Game — Where QMC Fails III

Exact Winning Probability (Formula 1.4.1)

Let $\rho(i_1, i_2)$ (rho (roe), the Greek letter ρ) denote the probability of winning given that the game starts at state (i_1, i_2). A closed-form formula is:

$$\rho(i_1, i_2) = \frac{\prod_{j=1}^{2} \left(\sum_{n_j=1}^{i_j} \prod_{r=1}^{n_j-1} \frac{q_j(r)}{p_j(r)} \right)}{\prod_{j=1}^{2} \left(\sum_{n_j=1}^{N_j} \prod_{r=1}^{n_j-1} \frac{q_j(r)}{p_j(r)} \right)}. \tag{1.4.1}$$

1.6.2 Example 2: A 2-D Game — Where QMC Fails IV

Notation for the Winning Probability Formula

In this formula:

- $\rho(i_1, i_2)$ (rho of i_1, i_2) is the probability of winning from starting position (i_1, i_2), a number in $[0, 1]$.
- $\prod_{j=1}^{2}$ (Pi from j equals 1 to 2) multiplies together the two terms for $j = 1$ and $j = 2$.
- $\sum_{n_j=1}^{i_j}$ (Sigma from n_j equals 1 to i_j) sums over all intermediate positions from 1 to the starting coordinate i_j.
- $\prod_{r=1}^{n_j-1}$ (Pi from r equals 1 to $n_j - 1$) multiplies the probability ratios along the path.
- $q_j(r)/p_j(r)$ is the ratio of the step-down probability to the step-up probability at position r along axis j. For a symmetric game ($p_j = q_j$), this ratio equals 1 everywhere.
- Empty products (when $n_j = 1$) equal 1 by convention.

1.6.2 Example 2: A 2-D Game — Where QMC Fails V

The symmetric special case. Set $N_1 = N_2 = 4$ and $p_j(i_j) = q_j(i_j) = 1/4$ for all positions $i_j \in \{1, 2, 3\}$. Then all ratios $q_j(r)/p_j(r) = 1$, all inner products equal 1, and all sums simplify. The formula reduces to $\rho(i_1, i_2) = (i_1/N_1)(i_2/N_2)$. Starting at $(i_1, i_2) = (2, 3)$:

$$\rho(2, 3) = \frac{2}{4} \cdot \frac{3}{4} = \frac{3}{8} = 0.375.$$

This is the *exact* winning probability that our simulation should recover. Any estimator that converges to a value other than 0.375 is *wrong*.

1.6.2 Example 2: A 2-D Game — Where QMC Fails VI

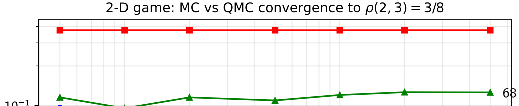
Simulation algorithm. To estimate $\rho(2,3)$, we play the game R times starting from $(2,3)$ and record $Y_j \in \{0,1\}$ (0 for LOSE, 1 for WIN) for each game $j = 1, \ldots, R$. The Monte Carlo estimator is:

$$\hat{Y}_R = \frac{1}{R} \sum_{j=1}^{R} Y_j,$$

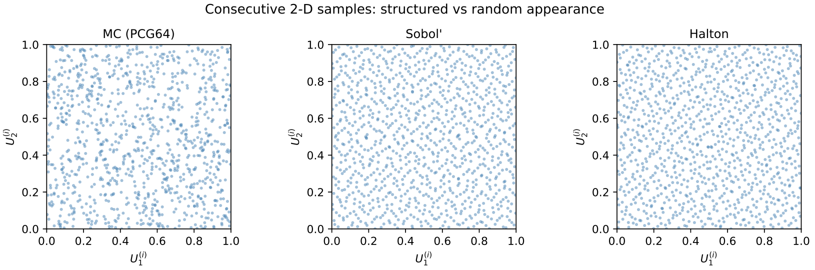
the fraction of games that end in WIN. Each *step* of the game uses a pair of uniform numbers $(U_1, U_2) \in [0,1)^2$: one to decide which axis to move along and one to decide the direction.

Why QMC fails here. Each game requires a *variable and random* number of (U_1, U_2) pairs from the uniform stream, depending on how long the game lasts before hitting a boundary. Game j's duration (measured in steps) is itself random and depends on the previous random outcomes. After game j finishes, the position in the QMC stream is at a random and unpredictable location. The structured correlations of the QMC sequence — designed for a fixed, predetermined access pattern — are completely disrupted when the stream is accessed at random, varying positions. This creates systematic correlations between games and causes *systematic bias* in the estimator.

Results (book Table 1.2, $R = 5000$ games)



1.6.2 Example 2: A 2-D Game — Where QMC Fails VII



Scatter plots of consecutive uniform pairs $(U_1^{(i)}, U_2^{(i)})$ drawn from $n = 1024$ samples. Left (MC / PCG64): no visible structure — the pairs look like random scatter. Centre (Sobol') and Right (Halton): clear linear patterns (points lie along diagonal stripes). When a game's stream position depends on previous random outcomes, these structured patterns create correlations between successive games and introduce systematic bias in the estimator.

Practical Rule: When to Use QMC and When Not To

Use QMC for **deterministic quadrature**: situations where a fixed, predetermined number of function evaluations is performed at points chosen before the computation begins. Do **not** use QMC for any simulation where the number of random numbers consumed per trial is itself random (as in any Markov chain simulation, sequential decision process, or stochastic simulation with variable run lengths). Standard MC with a high-quality PRNG is the safe and correct choice for all such tasks. The bias introduced by QMC in such settings cannot be corrected by simply using more samples — the estimator converges to the wrong value.

1.6.3 Python: 2-D Game Simulation (MC Version)

The following code simulates the symmetric 2-D game ($N_1 = N_2 = 4$, $p_j = q_j = 1/4$ for all positions) using standard pseudo-random numbers from PCG64. This correctly converges to the true value $\rho(2,3) = 0.375$. The key function `play_one_game` draws consecutive pairs (U_1, U_2) from the shared generator and uses them to determine each move.

```
import numpy as np
from numpy.random import default_rng

N1, N2 = 4, 4          # grid dimensions
p = q  = 0.25          # symmetric: p_j(i_j) = q_j(i_j) = 1/4 for all i_j

def play_one_game(rng, start=(2, 3), max_steps=10_000):
    """Play one 2-D game using consecutive (U1, U2) pairs from rng.

    At each step:
      U1 in [0, 0.5) -> change coordinate 1
      U1 in [0.5, 1) -> change coordinate 2
      U2 < 0.5       -> move UP (+1) in the chosen coordinate
      U2 >= 0.5      -> move DOWN (-1) in the chosen coordinate

    Returns 1 if WIN (both coords reach N), 0 if LOSE (any coord hits 0).
    """
    i1, i2 = start
    for _ in range(max_steps):
        U1, U2 = rng.random(), rng.random()
        if U1 < 2*p:                   # change coordinate 1 (prob = p + q = 0.5)
```

1.7 Bibliographical Notes I

Origins of Monte Carlo. The modern Monte Carlo method was developed at the Los Alamos National Laboratory in the late 1940s. Stanislaw Ulam conceived the basic idea while recovering from illness and playing solitaire: he realised that estimating probabilities by repeated random sampling could solve neutron diffusion equations that were intractable analytically. John von Neumann joined the effort and developed the theoretical foundations; Nicholas Metropolis implemented the method on early computers (the MANIAC machine at Los Alamos) and coined the name "Monte Carlo." The first published paper naming the method is: Metropolis & Ulam (1949), *The Monte Carlo Method*, *Journal of the American Statistical Association*, **44**(247):335–341.

Pseudorandom number generation. The foundational reference is D. E. Knuth [75], *The Art of Computer Programming*, Vol. 2, Chapter 3, which provides a comprehensive treatment of linear congruential generators and statistical tests for randomness. L'Ecuyer [82] surveys modern PRNGs with long periods and good statistical properties, including the Mersenne Twister. Niederreiter [101] covers both PRNGs and quasi-random sequences. The PCG64 generator used in NumPy was proposed by M. E. O'Neill (2014): *PCG: A Family of Simple Fast Space-Efficient Statistically Good Algorithms for Random Number Generation*, Preprint, Harvey Mudd College. PCG64 has period 2^{128} and excellent statistical quality; it is the recommended choice for Monte Carlo simulation in Python.

1.7 Bibliographical Notes II

Random walks and limit theorems. The arcsine law was established by W. Feller [52], *An Introduction to Probability Theory and Its Applications*, Vol. I, Chapters III and XIV. This book remains the most accessible and complete treatment of classical random walk theory and is essential reading for anyone working in probability. The Law of the Iterated Logarithm is due to A. Khintchine (1924) and A. N. Kolmogorov (1929); a modern probabilistic treatment is in Gut [100]. R. Durrett [43], *Probability: Theory and Examples* (4th edition, Cambridge University Press), provides a rigorous treatment using martingales and Brownian motion. The connection between the arcsine law and randomness testing of PRNGs is discussed in L'Ecuyer & Simard [82].

Quasi-Monte Carlo methods. The foundational monograph is H. Niederreiter [101], *Random Number Generation and Quasi-Monte Carlo Methods* (SIAM, 1992), which establishes the theory of low-discrepancy sequences, the Koksma–Hlawka inequality, and construction of the Halton, Faure, and Sobol' sequences. The Halton sequence was introduced in J. H. Halton [60] (1960); the Sobol' sequence in I. M. Sobol' [119] (1967); the Faure sequence in H. Faure [50] (1982). For scrambling and randomisation of QMC sequences, see S. Tezuka [124]. For financial applications of QMC, see P. Glasserman [55], *Monte Carlo Methods in Financial Engineering* (Springer, 2003). In Python, the

1.7 Bibliographical Notes III

`scipy.stats.qmc` module (SciPy $\geq$ 1.7) provides `Sobol`, `Halton`, and `LatinHypercube` classes with optional scrambling.

Travelling Salesman Problem. The TSP is studied exhaustively in D. L. Applegate, R. E. Bixby, V. Chvátal & W. J. Cook [4], *The Traveling Salesman Problem: A Computational Study* (Princeton University Press, 2006). Simulated annealing for combinatorial optimisation was introduced by S. Kirkpatrick, C. D. Gelatt & M. P. Vecchi [74] (1983), *Optimization by Simulated Annealing*, *Science*, **220**(4598):671–680. The two-player game used to illustrate QMC failure is from Lefebvre, L'Ecuyer & Simard [129]; the exact formula for the winning probability is derived there using the theory of Markov chains with absorbing states.

Chapter 1 Summary I

Chapter 1 introduces Monte Carlo through three concrete motivating examples — the simple symmetric random walk, the Travelling Salesman Problem, and numerical integration — and contrasts standard Monte Carlo with quasi-Monte Carlo for numerical integration. Each example was chosen to highlight a different strength of the Monte Carlo approach.

Core Concepts from Chapter 1

1. **The Monte Carlo estimator:** $\hat{Y}_n = \frac{1}{n}\sum_{i=1}^{n} k(\mathbf{U}_i)$ (Y-hat sub n, the sample average of n evaluations of k at uniform random points $\mathbf{U}_i$) is:
   - **Unbiased**: $\mathbb{E}[\hat{Y}_n] = I$ for all n, meaning it is correct on average.
   - **Consistent**: $\hat{Y}_n \to I$ almost surely by the SLLN (Strong Law of Large Numbers).
   - **Dimension-independent**: RMSE $= \sigma/\sqrt{n} = O(n^{-1/2})$ regardless of the dimension d.
2. **Simple symmetric random walk** $S_n = \sum_{j=1}^{n} X_j$ (S sub n, the sum of n i.i.d. steps $X_j \in \{-1, +1\}$):
   - The **Law of the Iterated Logarithm** (LIL): S_n stays almost surely in the envelope $\pm\sqrt{2n \log\log n}$, which grows slightly faster than $\sqrt{n}$.
   - The **Arcsine Law**: the fraction $L_{2n}^{+}/(2n)$ of time player A is winning follows a U-shaped arcsine distribution, not a distribution concentrated near $1/2$ as intuition would suggest.
3. **Travelling Salesman Problem (TSP)**: NP-hard with $n!$ possible tours; Monte Carlo heuristics (random restart, simulated annealing, cross-entropy method) find near-optimal tours far more efficiently than any exhaustive search.
4. **Discrepancy**: Measures the deviation of a point set from perfect uniformity. For i.i.d. uniform points: $D = O(n^{-1/2})$; for low-discrepancy (quasi-random) sequences: $D = O((\log n)^d/n)$.